TRIPATHI STUDY ZONE
JEE & NEET
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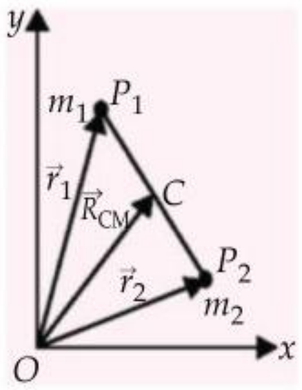
Centre of mass

The centre of mass of a system of particles is the point that moves as though

- All of the system mass is concentrated there and
- All external forces are applied there.

Centre of mass of a two particle system

- Consider a system of two particles P_1 and P_2 of masses m_1 and m_2 respectively. Let $\vec{r}_1$ and $\vec{r}_2$ be their position vectors with respect to the origin O, as shown in Fig. The position vector $\vec{R}_{CM}$ of the centre of mass C of the two particle system is given by



$$\vec{R}_{CM} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1 + m_2}$$

If $m_1 = m_2 = m$(say), then, $\vec{R}_{CM} = \frac{\vec{r}_1 + \vec{r}_2}{2}$

Thus the centre of mass of two equal masses lies exactly at the centre of the line joining the two masses.

If (x_1, y_1) and (x_2, y_2) are the coordinates of the locations of the two particles, then the coordinates of their centre of mass are given by

$$X_{CM} = \frac{m_1x_1 + m_2x_2}{m_1 + m_2} \text{ and } Y_{CM} = \frac{m_1y_1 + m_2y_2}{m_1 + m_2}$$

- In figure (a), two particles of masses m_1 and m_2 are separated by distance d. The position of the centre of mass (CM) of these two particles is given by



(a)

$$x_{CM} = \frac{m_2}{m_1 + m_2}d$$

Centre of mass of *n*-particles system

- For a system of n-particles of masses m_1, m_2, m_3, ..., m_n having the position vectors $\vec{r}_1, \vec{r}_2, ..., \vec{r}_n$ respectively with respect to coordinate system, the position of the centre of mass is given by a position vector

$$\vec{R}_{CM} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + ... + m_n\vec{r}_n}{m_1 + m_2 + ... + m_n} = \frac{\sum_{i=1}^{n} m_i\vec{r}_i}{M}$$

where, M is the total mass of the system.

The coordinates of centre of mass are given by

$$x_{CM} = \frac{\sum_{i=1}^{n} m_ix_i}{M} \; ; \; y_{CM} = \frac{\sum_{i=1}^{n} m_iy_i}{M} \; ; \; z_{CM} = \frac{\sum_{i=1}^{n} m_iz_i}{M}$$

Centre of mass of a rigid body (or continuous distribution of mass)

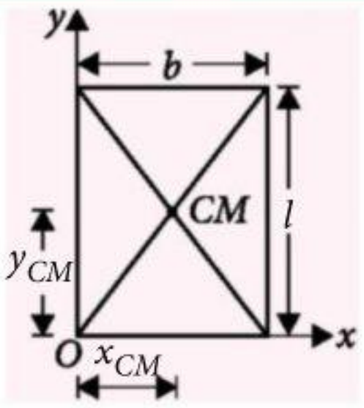
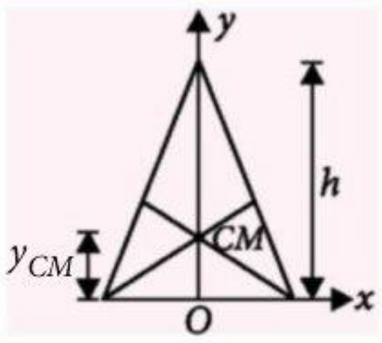
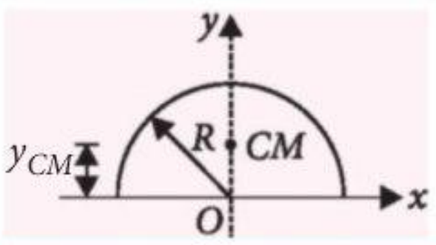
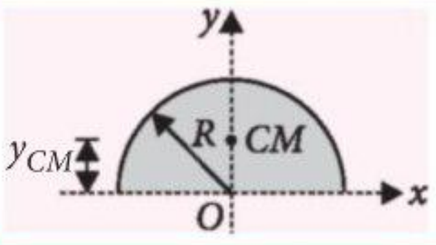
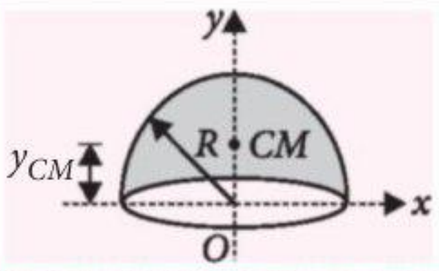
- The solid bodies contain so many particles (atoms) that we can treat the bodies as a continuous distribution of matter.

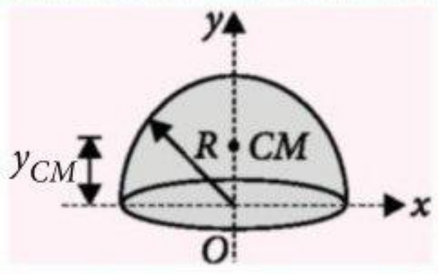
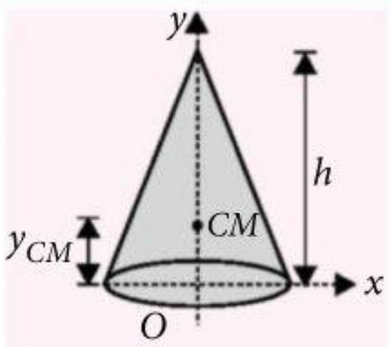
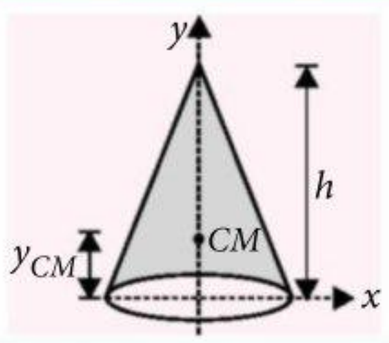
- The 'particles' then become differential mass elements *dm*, the sum become integrals, and the coordinates of the centre of mass are defined as

$$x_{CM}=\frac{1}{M}\int x\,dm\;;\; y_{CM}=\frac{1}{M}\int y\,dm\;;\; z_{CM}=\frac{1}{M}\int z\,dm$$

where M is the mass of the body.

- The position of the centre of mass of a system is independent of the choice of coordinate system.
- The position of the centre of mass depends on the shape of the body and the distribution of its mass. Hence it may lie within or outside the material of the body.
- In symmetrical bodies in which the distribution of mass is homogeneous, the centre of mass coincides with the centre of symmetry or geometrical centre.
- **Centre of mass of some well known symmetric rigid bodies :**

Position of Centre of Mass of different bodies	
	Rectangular plate $x_{CM}=\frac{b}{2}$, $y_{CM}=\frac{l}{2}$
	Triangular plate At the centroid, $y_{CM}=\frac{h}{3}, x_{CM}=0$
	Semi-circular ring $y_{CM}=\frac{2R}{\pi}, x_{CM}=0$
	Semi-circular disc $y_{CM}=\frac{4R}{3\pi}, x_{CM}=0$
	Hemispherical shell $y_{CM}=\frac{R}{2}, x_{CM}=0$

	Solid hemisphere $y_{CM}=\frac{3R}{8}, x_{CM}=0$
	Circular cone (solid) $y_{CM}=\frac{h}{4}, x_{CM}=0$
	Circular cone (hollow) $y_{CM}=\frac{h}{3}, x_{CM}=0$

Motion of Centre of Mass

- For a system of particles, position of centre of mass is $\vec{R}_{CM}=\frac{m_1\vec{r}_1+m_2\vec{r}_2+m_3\vec{r}_3+.....}{m_1+m_2+m_3+.....}$

  So, Velocity of centre of mass

$$\vec{v}_{CM}=\frac{m_1\vec{v}_1+m_2\vec{v}_2+.....}{m_1+m_2+.....}\qquad\left(\because\ \frac{d\vec{r}}{dt}=\vec{v}\right)$$

- Acceleration of centre of mass

$$\vec{a}_{CM}=\frac{m_1\vec{a}_1+m_2\vec{a}_2+....}{m_1+m_2+....}\qquad\left(\because\ \vec{a}=\frac{d\vec{v}}{dt}\right)$$

- Linear momentum of a system of particles is equal to the product of mass of the system with the velocity of its centre of mass.
- If no external force acts on a system the velocity of its centre of mass remains constant, *i.e.*, velocity of centre of mass is unaffected by internal forces.

  From Newton's second law $\vec{F}_{ext}=\frac{d(M\vec{v}_{CM})}{dt}$

  If $\vec{F}_{ext}=0$ then $\vec{v}_{CM}=$ constant.

Angular velocity and Angular acceleration

Angular Velocity

- It is defined as the time rate of change of angular displacement and is given by $\omega=\frac{d\theta}{dt}$.

- Angular velocity is directed along the axis of rotation. Angular velocity is a vector quantity. Its SI unit is rad s^{-1} and its dimensional formula is $[M^0L^0T^{-1}]$.

Relationship between Linear velocity and Angular Velocity

- The linear velocity of a particle of a rigid body rotating about a fixed axis is given by $\vec{v} = \vec{\omega} \times \vec{r}$ where $\vec{r}$ is the position vector of the particle with respect to an origin along the fixed axis.
- As in pure translation motion, all particles of the body have the same linear velocity at any instant of time, in pure rotational motion, all particles of the body have the same angular velocity at any instant of time.
- For rotation about a fixed axis, the direction of the angular velocity ω does not change with time. Its magnitude, however may change from instant to instant. In general rotation, both the magnitude and direction of ω may change from instant to instant.

Angular Acceleration

- It is defined as the time rate of change of angular velocity and is given by, $\alpha = \frac{d\omega}{dt}$
- Angular acceleration is a vector quantity. Its SI unit is rad s^{-2} and its dimensional formula is $[M^0L^0T^{-2}]$

Kinematics Equations of Rotational Motion

- $\omega = \omega_0 + \alpha t$
- $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$
- $\omega^2 = \omega^2_0 + 2\alpha\theta$

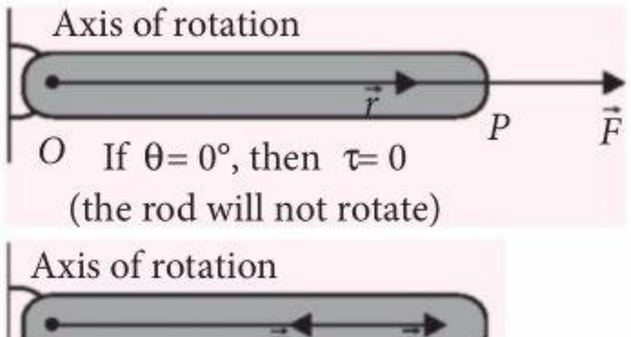
where the symbols have their usual meanings.

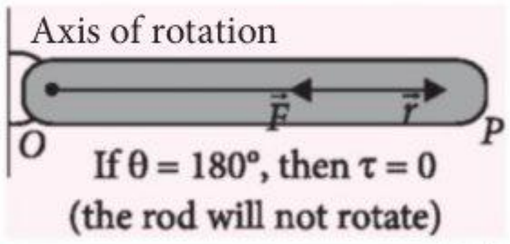
Torque and Angular Momentum

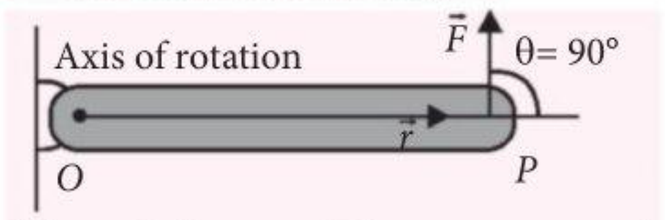
Torque

- Torque is the turning effect of a force. If a force acting on an object has a tendency to rotate the body about an axis, the force is said to exert a torque on the body. It is a vector quantity.
- In vector form,
  Torque, $\tau = \vec{r} \times \vec{F}$
  In magnitude, $\tau = r F \sin\theta$.
  Here θ is the angle between $\vec{r}$ and $\vec{F}$.
- Torque has the same dimensions as that of work *i.e.* $[ML^2T^{-2}]$. But work is a scalar quantity whereas torque is a vector quantity.
- By convention, anticlockwise moments are taken as positive and clockwise moments are taken as negative.

Special Cases

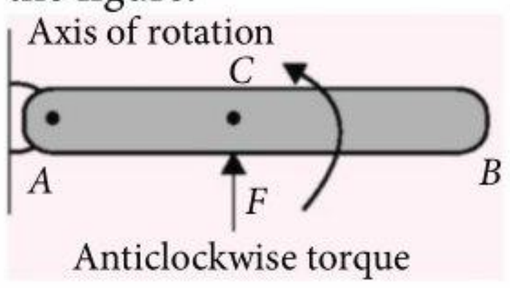
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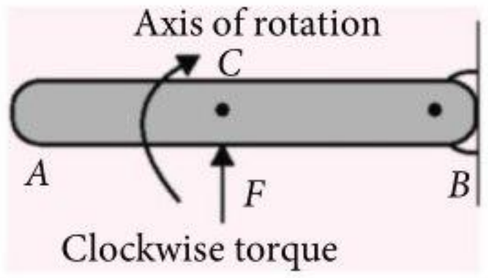
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If θ = 90°, then τ= *rF* (maximum torque)

Note : Same force acting at the same point can produce either anticlockwise or clockwise torque depending upon the location of the axis of rotation as shown in the figure.





Work Done by a Torque and Power

- Work done, W = torque × angular displacement $= \tau \times \Delta\theta$
- Power, $P = \frac{dW}{dt} = \tau\frac{d\theta}{dt} = \tau\omega$

Angular Momentum

- Angular momentum of a particle about a given axis is the moment of linear momentum of the particle about that axis. It is denoted by symbol $\vec{L}$.
- Angular momentum $\vec{L} = \vec{r} \times \vec{p}$
  In magnitude, $L = rp\sin\theta$
  where θ is the angle between $\vec{r}$ and $\vec{p}$.
- Angular momentum is a vector quantity. Its SI unit is kg m^2 s^{-1}. Its dimensional formula is $[ML^2T^{-1}]$.

Relationship between Torque and Angular Momentum

Rate of change of angular momentum of a body is equal to the external torque acting upon the body.

Mathematically, $\vec{\tau}_{ext} = \frac{d\vec{L}}{dt}$

Law of Conservation of Angular Momentum

If no external torque acts on a system then the total angular momentum is conserved.

$$i.e.,\ \vec{\tau}_{ext}=0 \ \text{ then } \frac{d\vec{L}}{dt}=0 \text{ or } \vec{L}=\text{constant}$$

Thus, if no external torque acts on a system, total angular momentum of the system remains constant. This is a statement of law of conservation of angular momentum.

Applications of Conservation of Angular Momentum

- A planet revolves around the sun in an elliptical path. When it comes near the sun, the moment of inertia of the planet about the sun decreases. In order to conserve the angular momentum, the angular velocity shall increase. Similarly, when the planet is away from the sun, there will be decrease in the angular velocity.
- An ice-skater or a ballet dancer can increase her angular velocity by folding her arms and bringing the stretched leg close to the other leg. When she stretches her hands and a leg outward, her moment of inertia increases and hence angular speed decreases to conserve angular momentum. When she folds her arms and brings the stretched leg close to the other leg, her moment of inertia decreases and hence angular speed increases.

Equilibrium of a Rigid Body

- A body is in rotational equilibrium, if the total external torque on it is zero, *i.e.* $\Sigma\tau_i = 0$.

  $\frac{d\vec{L}}{dt}=0 \Rightarrow \vec{L}=\text{constant}$

  i.e., the body rotates with a constant angular velocity.
- **Center of gravity :** A point of an extended body where the weight of the body acts and total gravitational torque on the body is zero, is known as the centre of gravity.

Peep Into Previous Years

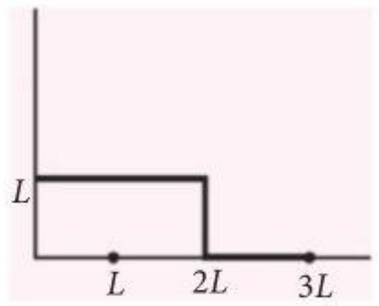
1. The position vector of the centre of mass $\vec{r}_{cm}$ of an asymmetric uniform bar of negligible area of cross-section as shown in figure is

(a) $\vec{r}_{cm}=\frac{11}{8}L\hat{x}+\frac{3}{8}L\hat{y}$

(b) $\vec{r}_{cm}=\frac{13}{8}L\hat{x}+\frac{5}{8}L\hat{y}$

(c) $\vec{r}_{cm}=\frac{3}{8}L\hat{x}+\frac{11}{8}L\hat{y}$

(d) $\vec{r}_{cm}=\frac{5}{8}L\hat{x}+\frac{13}{8}L\hat{y}$



(JEE Main 2019)

2. The moment of the force, $\vec{F}=4\hat{i}+5\hat{j}-6\hat{k}$ at (2, 0, –3), about the point (2, –2, –2), is given by

(a) $-8\hat{i}-4\hat{j}-7\hat{k}$ (b) $-4\hat{i}-\hat{j}-8\hat{k}$

(c) $-7\hat{i}-8\hat{j}-4\hat{k}$ (d) $-7\hat{i}-4\hat{j}-8\hat{k}$

(NEET 2019)

Moment of Inertia

- The moment of inertia of a rigid body about a fixed axis is defined as the sum of the products of the masses of the particles constituting the body and the squares of their respective distances from the axis of rotation.
- The dimensional formula of moment of inertia is $[ML^2T^0]$. The SI unit of moment of inertia is kg m^2 and its CGS unit is g cm^2.
- Momentum of inertia is the rotational analogue of mass in linear motion.

  The moment of inertia of a body depends on
  - Mass of the body
  - Size and shape of the body
  - Distribution of mass about the axis of rotation.
  - Position and orientation of the axis of rotation with respect to the body.

Radius of Gyration

- The radius of gyration of a body about its axis of rotation may be defined as the distance from the axis of rotation at which, if the whole mass of the body were concentrated, its moment of inertia about the given axis would be the same as with the actual distribution of mass. It is denoted by symbol k.
- The relation between moment of inertia I and radius of gyration k is

  $I=Mk^2 \text{ or } k=\sqrt{\frac{I}{M}}$
- Radius of gyration of a body about an axis of rotation may also be defined as the root mean square distance of its particles from the axis of rotation.

$$i.e.,\ k = \sqrt{\frac{r_1^2 + r_2^2 + ... + r_n^2}{n}}$$

The SI unit of radius of gyration is m.

Theorem of Perpendicular Axes

- It states that the moment of inertia of a plane lamina about an axis perpendicular to its plane is equal to the sum of the moments of inertia of the lamina about any two mutually perpendicular axes in its plane and intersecting each other at the point, where the perpendicular axes pass through the lamina. Mathematically,

$$I_z = I_x + I_y$$

where x-and y-axes lie in the plane of the lamina and z-axis is perpendicular to its plane and passes through the point of intersection of x and y-axes.

Theorem of Parallel Axes

- It states that the moment of inertia of a body about any axis is equal to the sum of the moment of inertia of the body about a parallel axis passing through its centre of mass and the product of its mass and the square of the distance between the two parallel axes. Mathematically, $I = I_{CM} + Md^2$

where I_{CM} is the moment of inertia of the body about an axis passing through its centre of mass and d is the perpendicular distance between the two parallel axes.

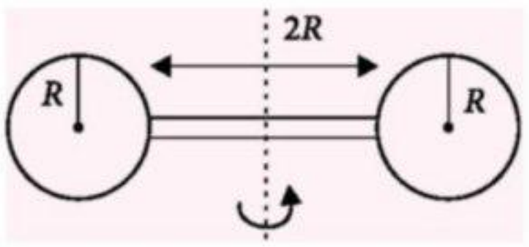
Moment of inertia and radius of gyration of some regular bodies about specific axis is given below :

S.No.	Body	Axis of rotation	Moment of inertia (I)	Radius of gyration (K)
1.	Uniform circular ring of mass M and radius R	(i) about an axis passing through the centre and perpendicular to its plane	MR^2	R
		(ii) about a diameter	$\frac{1}{2}MR^2$	$\frac{R}{\sqrt{2}}$
		(iii) about a tangent in its own plane	$\frac{3}{2}MR^2$	$\sqrt{\frac{3}{2}}R$
		(iv) about a tangent perpendicular to its plane	$2MR^2$	$R\sqrt{2}$
2.	Uniform circular disc of mass M and radius R	(i) about an axis passing through its centre and perpendicular to its plane	$\frac{1}{2}MR^2$	$\frac{R}{\sqrt{2}}$
		(ii) about a diameter	$\frac{1}{4}MR^2$	$R/2$
		(iii) about a tangent in its own plane	$\frac{5}{4}MR^2$	$\sqrt{5}\frac{R}{2}$
		(iv) about a tangent perpendicular to its own plane	$\frac{3}{2}MR^2$	$\sqrt{\frac{3}{2}}R$
3.	Solid sphere of radius R and mass M	(i) about its diameter	$\frac{2}{5}MR^2$	$\sqrt{\frac{2}{5}}R$
		(ii) about a tangential axis	$\frac{7}{5}MR^2$	$\sqrt{\frac{7}{5}}R$
4.	Hollow sphere of radius R and mass M	(i) about its diameter	$\frac{2}{3}MR^2$	$\sqrt{\frac{2}{3}}R$
		(ii) about a tangential axis	$\frac{5}{3}MR^2$	$\sqrt{\frac{5}{3}}R$

5.	Solid cylinder of length l, radius R and mass M	(i) about its own axis	$\frac{1}{2}MR^2$	$\frac{R}{\sqrt{2}}$
		(ii) about an axis passing through centre of mass and perpendicular to its own axis	$M\left[\frac{l^2}{12}+\frac{R^2}{4}\right]$	$\sqrt{\frac{l^2}{12}+\frac{R^2}{4}}$
		(iii) about the diameter of one of the faces of cylinder	$M\left[\frac{l^2}{3}+\frac{R^2}{4}\right]$	$\sqrt{\frac{l^2}{3}+\frac{R^2}{4}}$
6.	Hollow cylinder of mass M, length l and radius R	(i) about its own axis	MR^2	R
		(ii) about an axis passing through its centre of mass and perpendicular to its own axis	$M\left(\frac{R^2}{2}+\frac{l^2}{12}\right)$	$\sqrt{\frac{R^2}{2}+\frac{l^2}{12}}$
7.	Thin rod of length L and Mass M	(i) about an axis passing through the centre of mass and perpendicular to the rod	$\frac{ML^2}{12}$	$\frac{L}{\sqrt{12}}$
		(ii) about an axis passing through one end and perpendicular to rod	$\frac{ML^2}{3}$	$\frac{L}{\sqrt{3}}$
8.	Rectangular lamina of length l breadth b and mass M	about an axis passing through its centre of mass and perpendicular to plane	$M\left[\frac{l^2+b^2}{12}\right]$	$\sqrt{\frac{l^2+b^2}{12}}$
9.	Uniform solid cone of radius R, and mass M height h	about an axis through its centre of mass and joining its vertex to centre of base	$\frac{3}{10}MR^2$	$R\sqrt{\frac{3}{10}}$
10.	Parallelopiped of length l, breadth b and height h, mass M	about its central axis	$M\left(\frac{l^2+b^2}{12}\right)$	$\sqrt{\frac{l^2+b^2}{12}}$

Peep Into Previous Years

3. Two identical spherical balls of mass M and radius R each are stuck on two ends of a rod of length $2R$ and mass M(see figure). The moment of inertia of the system about the axis passing perpendicularly through the centre of the rod is



(a) $\frac{209}{15}MR^2$ (b) $\frac{137}{15}MR^2$

(c) $\frac{152}{15}MR^2$ (d) $\frac{17}{15}MR^2$

(JEE Main 2019)

4. A solid cylinder of mass 2 kg and radius 4 cm is rotating about its axis at the rate of 3 rpm. The torque required to stop it after 2π revolutions is

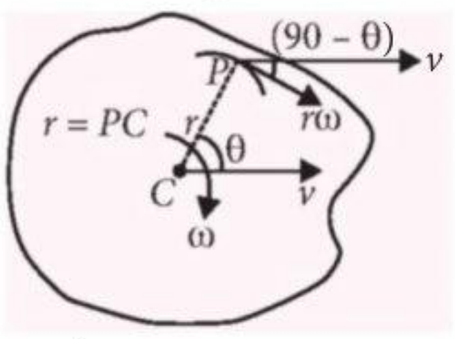
(a) 2×10^{6} N m (b) 2×10^{-6} N m

(c) 2×10^{-3} N m (d) 12×10^{-4} N m

(NEET 2019)

Rolling

- Rolling is a special phenomenon and it can be understood from two perspectives. Firstly, it can be seen as a special combination of rotation and translation. Secondly, it can be seen as pure rotation about its point of contact.
- Here, velocity of point P is the vector sum of two terms v and $r\omega$. Here v is common for all points, while $r\omega$ is different for different points, as r is different.

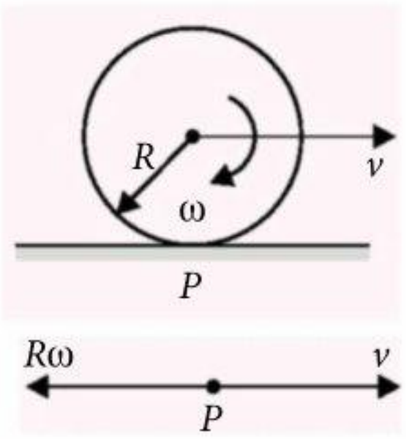


$$v_p = \sqrt{v^2 + (r\omega)^2 + 2(v)(r\omega)\cos(90° - \theta)}$$

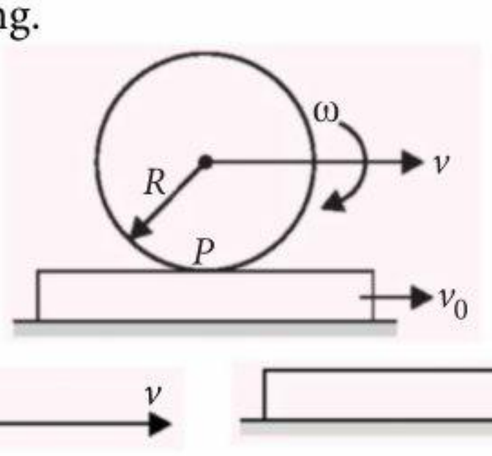
$$= \sqrt{v^2 + r^2\omega^2 + 2vr\omega \sin\theta}$$

- Uniform pure rolling : In which v and ω remain constant.

Condition of pure rolling is $v = R\omega$. In this case bottommost point of the spherical body is at rest. It has no slipping with its contact point on ground. Because ground point is also at rest.



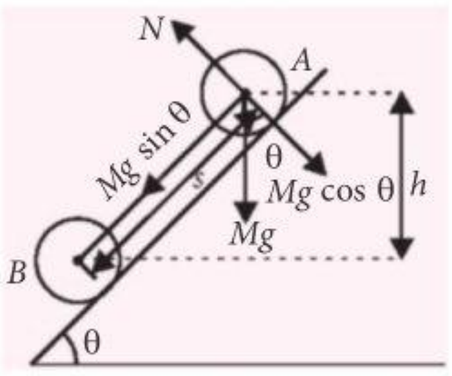
- If $v > R\omega$, then net velocity of point P is in the direction of v. This is called forward slipping.
- If $v < R\omega$, then net velocity of point P is in opposite direction of v. This is called backward slipping.



- If a spherical body is rolling over a plank, condition for no slipping between spherical body and plank is, $v - R\omega = v_0$

Motion of a Body Rolling without Slipping Down an Inclined Plane

- When a rigid body rolls down an inclined plane without slipping, it has a motion of translation as well as of rotation. As it rolls down, it suffers a vertical fall and therefore losses potential energy. At the same time it acquires linear and angular speeds and hence gains kinetic energy of translation and that of rotation. If there is no loss of energy through friction etc., the loss in gravitational potential energy is equal to the gain in kinetic energy.



- Consider a plane inclined at an angle θ to the horizontal and M be the mass and R the radius of that circular section of the body which rolls on the plane.
- Let initially the body be at point A and at rest and after some time it reaches B (figure.).
- In moving from point A to B, the body travels a vertical distance h given by $h = s \sin\theta$, and hence the loss of gravitational potential energy of the body = $Mgh = Mgs \sin\theta$
- Because there is no slipping and it is assumed that there is no dissipation of energy in any other way, the loss in potential energy must be equal to the gain in kinetic energy of the rolling body. Thus,

$$Mgs \sin\theta = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$$

Where I is moment of inertia of the body about an axis passing through its centre and perpendicular to its end faces. Now above equation can be written as

$$Mgs \sin\theta = \frac{1}{2}Mv^2 + \frac{1}{2}I\left(\frac{v^2}{R^2}\right) \quad ...(i)$$

On solving,

$$v = \sqrt{\frac{2Mgh}{M + I/R^2}} = \sqrt{\frac{2gh}{1 + \frac{k^2}{R^2}}}$$

- Differentiating equation (i), w.r.t. time, we get acceleration of rolling body,

$$a = \frac{Mg\sin\theta}{M + I/R^2} = \frac{g\sin\theta}{1 + \frac{k^2}{R^2}}$$

- Time of descent of a body rolling down an inclined plane, from equation $s = ut + \frac{1}{2}at^2$ will be,

$$t = \frac{1}{\sin\theta}\sqrt{\frac{2h}{gM}\left(M + \frac{I}{R^2}\right)} = \frac{1}{\sin\theta}\sqrt{\frac{2h}{g}\left(1 + \frac{k^2}{R^2}\right)}$$

Analogy between Translational Motion and Rotational Motion

The analogy between quantities that describe linear motion and the corresponding quantities that describe rotational motion is as shown in the table.

	Linear motion	Rotational motion about a fixed axis
1.	Displacement, x	Angular displacement, θ
2.	Velocity, $v = \frac{dx}{dt}$	Angular velocity, $\omega = \frac{d\theta}{dt}$

3.	Acceleration $a = \frac{dv}{dt}$	Angular acceleration $\alpha = \frac{d\omega}{dt}$
4.	Mass, M	Moment of inertia, I
5.	Force, $F = Ma$	Torque, $\tau = I\alpha$
6.	Work done, $W = Fs$	Work done, $W = \tau\theta$
7.	Translational kinetic energy, $K_T = \frac{Mv^2}{2}$	Rotational kinetic energy, $K_R = \frac{I\omega^2}{2}$
8.	Power, $P = Fv$	Power, $P = \tau\omega$
9.	Linear momentum, $p = Mv$	Angular momentum, $L = I\omega$
10.	Equations of translational motion $v = u + at$, $s = ut + \frac{1}{2}at^2$ $v^2 - u^2 = 2as$ where the symbols have their usual meaning.	Equations of rotational motion $\omega = \omega_0 + \alpha t$ $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$ $\omega^2 - \omega_0^2 = 2\alpha\theta$ where the symbols have their usual meaning.

Peep Into Previous Years

5. Two coaxial discs, having moments of inertia I_1 and $\frac{I_1}{2}$, are rotating with respective angular velocities ω_1 and $\frac{\omega_1}{2}$, about their common axis. They are brought in contact with each other and thereafter they rotate with a common angular velocity. If E_f and E_i are the final and initial total energies, then $(E_f - E_i)$ is

(a) $-\frac{I_1\omega_1^2}{24}$ (b) $\frac{3}{8}I_1\omega_1^2$

(c) $\frac{I_1\omega_1^2}{6}$ (d) $-\frac{I_1\omega_1^2}{12}$

(JEE Main 2019)

6. A solid sphere is in rolling motion. In rolling motion a body possesses translational kinetic energy (K_t) as well as rotational kinetic energy (K_r) simultaneously. The ratio $K_t : (K_t + K_r)$ for the sphere is

(a) 7 : 10 (b) 5 : 7

(c) 10 : 7 (d) 2 : 5 ***(NEET 2018)***

Points For Extra Scoring

- A point mass is tied to one end of a string which is wound around a solid body, then. When the mass is released, it falls vertically downwards. Downward acceleration of point mass,

$$a = \frac{g}{1 + \frac{I}{mR^2}}$$

Tension is string, $T = mg\left[\frac{I}{I + mR^2}\right]$

Velocity of point mass, $v = \sqrt{\frac{2gh}{1 + \frac{I}{mR^2}}}$

Angular velocity of rigid body, $\omega = \sqrt{\frac{2mgh}{I + mR^2}}$

where m is the mass of point mass, R is the radius of the rigid body and I is the moment of inertia of the rigid body.

- Moment of Inertia of annular disc through an axis

Passing through the centre and perpendicular to the plane

$$I = \frac{M}{2}[R_1^2 + R_2^2]$$

Passing through its diameter.

$I = \frac{M}{4}[R_1^2 + R_2^2]$, where R_1 and R_2 are the inner and outer radius respectively.

- Moment of inertia of an equilateral triangular lamina with side a passing through its centre of mass and perpendicular to the plane

$$I = \frac{Ma^2}{6}$$

Answer Key For Peep Into Previous Years

1.	(b)	2.	(d)	3.	(b)	4.	(b)	5.	(a)	6.	(b)